Name____________________

Three Dimensional Vector Operations

Date________________ Period____

Find the component form of the resultant vector.

1) $\vec{f} = \langle -6, -8, 8 \rangle$
$\vec{g} = \langle 8, -5, 3 \rangle$
Find: $\vec{f} + \vec{g}$

2) $\vec{u} = \langle 0, 9, -4 \rangle$
$\vec{g} = \langle 9, 7, -7 \rangle$
Find: $\vec{u} - \vec{g}$

3) Given: $A = (-6, -3, -5)$ $B = (-4, -8, 5)$
$C = (-2, 3, -4)$ $D = (-6, 5, 0)$
Find: $-\overrightarrow{AB} - \overrightarrow{CD}$

4) Given: $T = (-2, 4, -1)$ $X = (3, -2, -5)$
$Y = (7, 5, -1)$ $Z = (7, 3, -4)$
Find: $\overrightarrow{TX} + \overrightarrow{YZ}$

5) $\vec{a} = \langle 1, -8, 5 \rangle$
$\vec{g} = \langle -6, 7, -7 \rangle$
Find: $-\vec{a} - 4\vec{g}$

6) $\vec{u} = \langle -2, 5, -6 \rangle$
$\vec{g} = \langle -2, -3, 8 \rangle$
Find: $4\vec{u} + 10\vec{g}$

7) Given: $A = (-6, -6, -4)$ $B = (4, -8, 7)$
$C = (8, 9, 5)$ $D = (7, 2, -9)$
Find: $5\overrightarrow{AB} + 10\overrightarrow{CD}$

8) Given: $P = (0, 9, 7)$ $Q = (-9, -4, 1)$
$R = (9, -9, -4)$ $S = (-8, 2, 0)$
Find: $6\overrightarrow{PQ} - 5\overrightarrow{RS}$